EXPERIMENT NO - 2

AIM:

To Convert the given ER diagram into a relational schema and implement it using MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL Constraints. Also, to insert sample data and demonstrate referential Integrity violations.

Relational Schema:

CUSTOMER (Customer_ID, Name, Email, Phone)

ADDRESS (Address_ID, House_No, Street, City, State, PIN_Code, Customer_ID)

SELLER (Seller_ID, Seller_Name)

CATEGORY (Category_ID, Category_Name)

PRODUCT (product_ID, Name, Price, Seller_ID, Category_ID)

ELECTRONIC (Product_ID, Brand, Warranty)

CLOTHING (Product_ID, Size, Fabric)

ORDERS (Order_ID, Order_Date, Status, Customer_ID)

ORDER_ITEM (Order_ID, Product_ID, Quantity, Unit_Price)

PAYMENT (Payment_ID, Amount, Payment_Date, Method, Status, Order_ID)

DELIVERY (Delivery_ID, Delivery_Date, Status, Tracking_No, Order_ID)

Primary keys:

Customer_ID, Address_ID, Seller_ID, Category_ID,

Product_ID, Product_ID, Product_ID,

Order_ID (Order_ID, Product_ID),

Payment_ID, Delivery_ID.

Source Code :

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Name VARCHAR (50) NOT NULL,
    Email VARCHAR (100) NOT NULL UNIQUE,
    Phone VARCHAR (15) UNIQUE
);

CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    House_NO VARCHAR(20) NOT NULL,
    Street VARCHAR (100),
    City VARCHAR (50) NOT NULL,
    State VARCHAR (50) NOT NULL,
    PIN_Code VARCHAR (50) NOT NULL,
    Customer_ID INT,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE SET NULL
);

CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR (100) NOT NULL
);
```

```
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE Product (
    Product_ID IINT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10, 2) NOT NULL,
    Seller_ID INT NOT NULL,
    Category_ID INT,
    FOREIGN KEY (Seller_ID)
        REFERENCES Seller (Seller_ID)
        ON DELETE CASCADE,
    FOREIGN key (Category_ID)
        REFERENCES Category (Category_ID)
        ON DELETE SET NULL
);
```

Product Specialization + Order

```
CREATE TABLE Electronic (
    Product_ID INT PRIMARY KEY,
    Brand VARCHAR(50) NOT NULL,
    Warranty VARCHAR(50),
    FOREIGN KEY (Product_ID)
        REFERENCES Product (Product_ID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Clothing (
    Product_ID INT PRIMARY KEY,
    Size VARCHAR (10) NOT NULL,
    Fabric VARCHAR (50),
    FOREIGN KEY (Product_ID)
      REFERENCES Product (Product_ID)
     ON DELETE CASCADE
);
```

```
CREATE TABLE Clothing (
      Product_ID INT PRIMARY KEY,
      Size VARCHAR (10) NOT NULL,
      Fabric VARCHAR (50),
      FOREIGN KEY (Product_ID)
      ON DELETE CASCADE
);
```

```
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Order_Date DATE NOT NULL,
    Status VARCHAR (80) NOT NULL,
    Customer_ID INT NOT NULL,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Order_Item (
    Order_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL (10,2) NOT NULL,
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Product_ID)
        REFERENCES Product (Product_ID)
        ON DELETE CASCADE
);

CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Amount DECIMAL (10, 2) NOT NULL,
    Payment_Date DATE NOT NULL,
    Method VARCHAR (30) NOT NULL,
    Status VARCHAR (30) NOT NULL,
    Order_ID INT NOT NULL UNIQUE,

    FOREIGN_KEY (Order_ID)
        REFERENCES Order (Order_ID)
        ON DELETE CASCADE

);
```

```
CREATE Table Delivery (
    Delivery_ID INT PRIMARY KEY,
    Delivery_Date DATE,
    Status VARCHAR (30) NOT NULL,
    Tracking_No VARCHAR (50) UNIQUE,
    Order_ID INT NOT NULL UNIQUE,
    FOREIGN KEY (Order_ID)
      REFERENCES Order (Order_ID)
    ON DELETE CASCADE,
);

INSERT INTO Customer
VALUES
(1, 'Ananya', 'ananya@gmail.com', '9876543210'),
(2, 'Priya', 'priya@gmail.com', '9876543211');

INSERT INTO Address
VALUES
(101, '12A', 'MG Road', 'Lucknow', 'Uttar Pradesh', '226001'),
(102, '45B', 'Civil Lines', 'Prayagraj', 'Uttar Pradesh', '211002');

INSERT INO Seller
VALUES
(201, 'Tech World'),
(202, 'Fashion Hub');
```

```
INSERT INTO Category
VALUES
(301, 'electronics'),
(302, 'Clothing');

INSERT INTO Product
VALUES
(401, 'laptop', 55000.00, 201, 301),
(402, 'T-shirt', 999.00, 202, 302);

INSERT INTO Electronics
VALUES
(401, 'Dell', '2 years');

INSERT INTO Clothing
VALUES
(402, 'M', 'Cotton');

INSERT INTO Orders
VALUES
(501, '2026-08-20', 'Confirmed', 1);
(502, '2026-08-20', 'pending', 2);

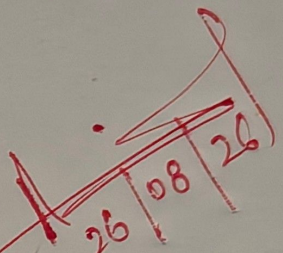
INSERT INTO Order_Item
VALUES
(501, 401, 1, 55000.00)
(502, 402, 2, 999.00);
```

```
INSERT INTO Payment
VALUES
(601, 55000.00, '2026-08-20', 'UPI', 'Paid', 501),
(602, 1998.00, '2026-08-20', 'Card', 'Paid', 502);

INSERT INTO DELIVERY
VALUES
(701, '2026-08-23', 'Shipped', 'TRK1001', 501),
(702, '2026-08-24', 'Processing', 'TRK1002', 502);
```

Referential Integrity violation

```
INSERT INTO Orders
VALUES
(503, '2026-08-20', 'Confirmed', 999);
```



ERROR: Cannot add or update a child row: a foreign key Constraint fails.

Referential violation Integrity - 2

```
INSERT INTO Order_Item
VALUES
(999, 401, 1, 55000.00)
```

ERROR: Cannot add or update a child row: a foreign key Constraint fails.

Output :-

The relational schema was Successfully Implemented using MYSQL CREATE TABLE statements. Sample records were Inserted Successfully, and referential Integrity Violations were demonstrated using Invalid foreign key Values.